

Pentium 4 ITLB, Branch Prediction & Hyper-Threading — MPCA Module 6

Definition: Pentium 4 integrates an **Instruction Translation Lookaside Buffer (ITLB)**, advanced **Dynamic Branch Prediction**, and **Hyper-Threading (HT) Technology** (Simultaneous Multithreading) to maximize execution unit utilization.

1. Detailed Technical Explanation

1. Instruction Translation Lookaside Buffer (ITLB)

- A high-speed associative hardware cache that caches virtual-to-physical page address translations for instruction fetch.
- Eliminates memory access latency to Page Tables in RAM during code prefetch.

2. Advanced Dynamic Branch Prediction

- Because the NetBurst pipeline is **20 stages deep**, a branch misprediction penalty results in flushing 20 stages of instructions (a massive performance penalty).
- NetBurst utilizes a **4 KB Branch Target Buffer (BTB)** combining:
- **Static Branch Predictor:** Assumes backward branches (loops) are TAKEN, forward conditional branches are NOT TAKEN.
- **Dynamic 2-Level Adaptive Branch Predictor:** Tracks global execution history patterns, achieving **>94% prediction accuracy**.

3. Hyper-Threading Technology (HT Technology)

Hyper-Threading is Intel's hardware implementation of **Simultaneous Multithreading (SMT)** that allows a single physical processor core to appear as **two logical processors** to the operating system.

WITHOUT HYPER-THREADING:

Physical Core: [Thread 1] ==> Idle Execution Units (Wasted Silicon)

WITH HYPER-THREADING (Simultaneous Multithreading):

Logical Core 1: [Thread 1] ==> Shared Physical Execution Units (ALU, FPU, Caches)

Logical Core 2: [Thread 2] / (Zero Idle Slots, 15-30% Throughput Boost!)

Architecture of Hyper-Threading:

1. Duplicated Architectural State:

- Registers (**EAX**, **EBX**, etc.), **EFLAGS**, Segment Selectors, Control Registers (**CR0-CR3**), and APIC interrupt

controllers are **duplicated** for each logical thread.

2. Shared Execution Resources:

- Execution Units (ALUs, FPUs), Trace Cache, L2 Cache, and Memory Subsystems are **shared dynamically** between the two threads.

3. Performance Benefit: Improves CPU execution pipeline utilization by up to **30%** with only a ~5% increase in physical silicon chip area.

4. Must-Write Points for Exams

- Hyper-Threading presents one physical core as two logical cores to the OS scheduler.
- Architectural state (GPRs, Flags, Program Counter) is duplicated; physical ALUs/FPUs/Caches are shared.
- Dynamic branch prediction accuracy (>94%) is vital to prevent severe 20-stage pipeline flush penalties.

5. Quick Recall Flow

ITLB Page Caching -> Advanced BTB (>94% Branch Accuracy) -> Hyper-Threading: 1 Physical Core = 2 Logical Processors (Duplicated State, Shared ALUs)